Module 2 Lab 1: Frame a Business Problem as a Machine Learning Problem

Submission template. 20 points. This lab has no notebook and no code. Fill this document in as you work, then upload it to Canvas. Check Canvas for the due date.

Your Unique Values

Your name	Justin Chang
Organization you chose in Step 1	Casino
GitHub portfolio repository URL	github.com/autoramen0
Date you started this lab	11/09/2026

Step 1: Your Business Request

Two sentences: what the organization is, and what it is asking for. Do not use the dental clinic example.

Organization and request	A Casino is looking to flag black jack card counters. Say the Casino has 60 tables, and analysts currently view video footage to catch card counters. Player play is currently compared to ideal play, and count, but the decision to back a player off must be made by a human. They're asking to design a tool that will flag people.

Prediction 1: Threshold

Write this BEFORE Step 3. Is the right cutoff for your problem closer to 0.05, 0.5, or 0.9? An honest wrong prediction earns full credit. Do not edit it afterward.

My predicted threshold	0.05
One sentence of reasoning	Depends on the bets that the player is making. Lets say on average 200$ per hour, and 14 hours before the player gets backed off on average.

Step 2: ML Problem

Fill in every line with something specific. No line may read depends or varies. This is a written form, not code.

Type (classification, regression, clustering, ranking, or none)	Classification
Input (one ____)	1 Player Session
Features (at least five)	Bet Spread: max bet/min bet Correlation: Bet size / True Count Rate of count based strategy deviations: Mid Shoe entry and exits Average bet and hands played per hour.
Output	0 to 1 probability someone is a counter
Decision (if ____ then ____)	Output > thresh-hold send the session to analyst for review.

Step 3: Threshold From Money

Show the arithmetic. If a figure is an estimate, label it as an estimate.

Cost of acting unnecessarily (false positive) $	25
Cost of missing a real case (false negative) $	400
Which figures are estimates, and how you estimated them	FP is an estimate: Looking at the data for BLS https://www.bls.gov/oes/2023/May/oes339031.htm analyst get's paid in vegas a mean of $22.69 /hr. With benefits, 1.35, and we assume it takes 40-50 minutes to make a judgement. FN is also an estimate: We assume that $B = \$100$ bet average, $\frac{dH}{dt} = 80$ hands per hour. $E = 0.01$ edge and $t = 5$ before a black jack player is flagged. Assumed that the Casino does not know about this player. Yields the following: $B \cdot \frac{dH}{dt} \cdot t \cdot E$ = FN Cost.
Arithmetic: p > FP cost / FN cost	$p > \frac{25}{400} = 0.0625$
Computed threshold	0.0625
Prediction 1 compared with the computed threshold	Pretty close I think. I just picked the lowest option since I would imagine the potential cost of the FP would be much

	lower than the FN. There is a reason why blackjack counters are backed off.

Step 4: The Metric

Metric I would optimize	Recall
Defense in two sentences, tied to which error costs more	Definitely the cost of a False Negative is much more than a False positive. It is imperative we identify actual cases of card counting than just a balanced approach. Costs next to nothing to have the analyst go through the tape when compared with potential losses.

Step 5: Data and Its Weakest Point

Labeled history required, roughly how much, and where it comes from	Ideally the data would be provided directly by the Casinos themselves. Several thousand records of individuals playing. Mix of people that were not backed off and people that were backed off.
The single weakest point in that data	How do you definitively define someone as a card counter. More-over, in the existing data sets what is the degree of False Negatives that exist. Are there people that human surveilance teams let escape? With the labels themselves they are not a true record of reality.

Prediction 2: The Deciding Property

Write this BEFORE Step 6. Which single property will decide the foundation model question: input modality, volume, latency, determinism, or explainability?

My predicted deciding property	Explainability

Step 6: Why or Why Not a Foundation Model

My conclusion (WHY or WHY NOT a foundation model)	We have to be able to justify why a player is backed off. Being able to have some sort of decision tree to refer to is important. More-over the domain is specific enough such that the pre-trained data of the foundation model may not be applicable to this business case.
The one property that decided it	I think explainability justifies it, with the idea of the specificity of the domain being a strong runner-up candidate. Maybe also determinism. You have to be able to apply to standard equally to given set of conditions. Where currently depending on whose working the shift and other conditions. This is a

	case that not only delves on blackjack being a game, but potential legal issues as well with regards to trespassing repeat offenders.
Prediction 2 compared with what actually decided it	Its match on

Reflection

In your own words, no AI assistance. Answer from your own numbers. An answer that could have been written without doing the lab earns no credit.

1. Report your Step 2 threshold prediction and your Step 3 computed threshold. If they differed, explain what your instinct was tracking that the money was not, or the reverse.	It was close enough. I think my estimates can be seen as conservative though. There's a wide degree of variability to this scenario. Some people who are already in a database don't even get to bet. Other times they may visit a casino, and it may take them a few hours before they get backed off. There's plenty of channels on youtube that kind of showcase people's card counting adventures with a hidden camera. Sometimes the tables themselves have limits on how much you can bet.
2. State the weakest point in your data and explain concretely how a model trained on it could look successful in testing and still fail in production.	I mentioned it in my analysis above. The labels are unreliable and arbitrary. There may very well be people who have never been backed off that are actually card-counting. That may be the case with someone that is not good at it, or maybe they are unlucky and get clobbered by variance. Although I do think the correlation with bet size and true-count provides a potential objective measure into whether or not someone is counting or not. Thinking about the different metrics too,
3. Someone senior tells you the threshold should be 0.5 because that is standard. Write the two sentence reply you would actually send, using your own numbers.	That decision is not defensible. For that to be the case each False negative would have to cost us 50$, which is true if the player isn't using particularly aggressive bets, to which they pose no problem to the casino. The thresh-hold must be sensitive because we are most concerned with the most aggressive and skilled card-counters.

AI Use Disclosure

List any AI tool you used, what you used it for, and how you checked what it produced. Write None if you used none. Predictions and reflection answers must be your own work with no AI assistance.

Tools used	Perplexity
What I used them for	BLS Data, and scoping out existing Casino Data. It turns out the data for this specific use case isn't widely available.

How I verified the output	Clicked on the sources in perplexity. Made sure that the claims were from the organizations that were stating it. Followed the chain if there was a reference on the source.

Before You Submit

- ☐ My threshold came from arithmetic, not from 0.5
- ☐ My data risk names something specific about how the label is produced
- ☐ Both predictions were written before I did the matching step
- ☐ I recorded the actual result for both predictions, including any I got wrong
- ☐ This template is pushed to my GitHub portfolio repository and the URL is above
- ☐ Reflection answers written by me with no AI assistance
- ☐ Files uploaded individually, no archive